Non-Simply Connected Subsystems and the Reduced Density Matrices

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Abstract

We discuss the structure of reduced density matrices of non-simply connected subsystems. The subtle difference in fermion-parity properties between the physical Kitaev model and its Majorana equivalence influences the structure of the reduced density matrix and the entanglement spectrum. As a result, although the block-diagonal organization remains, the detailed entanglement spectrum changes, and the Yao-Qi method must be modified.

1 Patch Configurations

For pedagogical reasons, we first consider a non-simply connected region A with one interior hole B to demonstrate a hidden structure that was not considered previously in simply connected subsystems. First, we have to take the non-contractible closed loop $\mathcal{L}$ inside the subsystem A encircling the interior hole B into account. The second subtle thing is the edge configuration $\mathfrak{H}$, which represents the *patch configuration* of the hole enclosed by A as illustrated in Fig. 1(a). Note that $\Sigma_{\mathfrak{D}\ominus\mathfrak{H}}$ is shorthand notation for the string operator of the combination of $\mathfrak{D}$ combined with the patch configuration, but with the local identity operators generated by the relation $\sigma^x\sigma^y\sigma^z \propto \mathbb{1}$ inside the hole B excluded. With these two additional parts, the reduced density matrix of this subsystem is as follows,

$$\rho_A = \frac{1}{2^{N^A - N_p^A}} \hat{\Pi}_A (\mathbb{1} + W_{\mathcal{L}}) \sum_{[\mathfrak{D}]\subset A} \left(\mathfrak{C}_{\mathfrak{D}} \Sigma_{\mathfrak{D}} + \mathfrak{C}_{\mathfrak{D}\ominus\mathfrak{H}} \Sigma_{\mathfrak{D}\ominus\mathfrak{H}} \right).$$

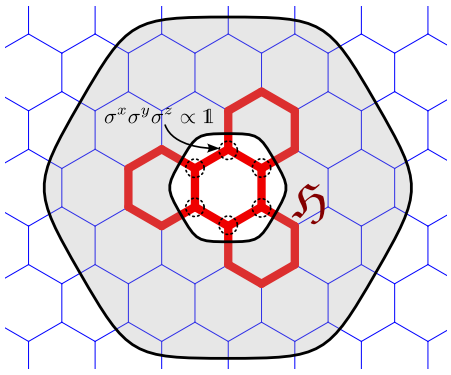


Figure 1: The patch configuration $\mathfrak{H}$ covering the interior hole of a non-simply connected subsystem. This configuration is devised in such a way that the corresponding string operator $\Sigma_{\mathfrak{H}}$ has all its local components inside the hole B equal to the identity. Its corresponding string operator $\Sigma_{\mathfrak{H}}$ commutes with other string operators lying within the shaded subsystem.

The non-contractible loop Wilson operator $W_{\mathcal{L}}$ with the identity forms a projector due to the 1-form Wilson symmetry. However, the expectation values of string operators $\Sigma_{\mathfrak{D}\ominus\mathfrak{H}}$ are not constant, and each operator $\Sigma_{\mathfrak{D}\ominus\mathfrak{H}}$ cannot be separated into $\Sigma_{\mathfrak{D}}$ and $\Sigma_{\mathfrak{H}}$ without phase factors either. As a result, the string operators containing the patch operator cannot be written as projectors, unlike the non-contractible loop $W_{\mathcal{L}}$. Therefore, we take the other way around to bypass this difficulty. We first decompose the string operator $\Sigma_{\mathfrak{D}\ominus\mathfrak{H}}$ into the form $\varphi(\mathfrak{H},\mathfrak{D})\Sigma_{\mathfrak{H}}\Sigma_{\mathfrak{D}}$, and then write the reduced density matrix as follows,

$$\rho_A = \frac{1}{2^{N^A-N_p^A-1}}\hat{\Pi}_A \underbrace{\left(\frac{\mathbb{1}+W_{\mathcal{L}}}{2}\right)}_{\equiv\hat{\Pi}_{\mathcal{L}}} \sum_{[\mathfrak{D}]\subset A} \left(\mathbb{1} + \underbrace{\varphi(\mathfrak{H},\mathfrak{D})\frac{\mathfrak{C}_{\mathfrak{D}\ominus\mathfrak{H}}}{\mathfrak{C}_{\mathfrak{D}}}}_{\equiv 2p_{\mathfrak{D}}^{(+)}-1}\Sigma_{\mathfrak{H}}\right)\mathfrak{C}_{\mathfrak{D}}\Sigma_{\mathfrak{D}}.$$

In the above equation, we defined a new quantity $p_{\mathfrak{D}}^{(+)}$ derived from the phase factor and the string coefficients. Applying the following relation

$$\frac{1+aA}{2} = \left(\frac{1+a}{2}\right)\left(\frac{1+A}{2}\right) + \left(\frac{1-a}{2}\right)\left(\frac{1-A}{2}\right),$$

the matter part of the density matrix can be decomposed. As a result, we obtain

$$\rho_A = \frac{1}{2^{N^A-N_p^A-2}}\hat{\Pi}_A\hat{\Pi}_{\mathcal{L}} \sum_{[\mathfrak{D}]\subset A}\left[p_{\mathfrak{D}}^{(+)}\left(\frac{\mathbb{1}+\Sigma_{\mathfrak{H}}}{2}\right) + p_{\mathfrak{D}}^{(-)}\left(\frac{\mathbb{1}-\Sigma_{\mathfrak{H}}}{2}\right)\right]\mathfrak{C}_{\mathfrak{D}}\Sigma_{\mathfrak{D}}$$

where $p_{\mathfrak{D}}^{(-)} \equiv 1 - p_{\mathfrak{D}}^{(+)}$ with $p_{\mathfrak{D}}^{(+)}$ defined at the beginning forms a binary probability distribution $\{p_{\mathfrak{D}}^{(+)}, p_{\mathfrak{D}}^{(-)}\}$ normalized as $p_{\mathfrak{D}}^{(+)} + p_{\mathfrak{D}}^{(-)} = 1$. The projectors $(\mathbb{1} \pm \Sigma_{\mathfrak{H}})/2$, which will be denoted by the shorthand notation $\hat{\Pi}_{\mathfrak{H}}^{(\pm)}$ later on, divide the local Hilbert space into two orthogonal sectors with the eigenvalues of $\Sigma_{\mathfrak{H}}$ being $+1$ and -1, respectively. As a result, the matter part of the reduced density matrix becomes a direct sum of the two sectors.

$$\rho_A \equiv \frac{1}{2^{N^A-N_p^A-2}}\hat{\Pi}_A\hat{\Pi}_{\mathcal{L}} \sum_{[\mathfrak{D}]\subset A}\mathfrak{C}_{\mathfrak{D}}\left[\left(p_{\mathfrak{D}}^{(+)}\hat{\Pi}_{\mathfrak{H}}^{(+)}\Sigma_{\mathfrak{D}}\right)\oplus\left(p_{\mathfrak{D}}^{(-)}\hat{\Pi}_{\mathfrak{H}}^{(-)}\Sigma_{\mathfrak{D}}\right)\right].$$

The reduced density matrix expressed in the above form implies that the coefficients $p_{\mathfrak{D}}^{(s)}$ represent the probability of finding Majorana fermions in each sector labeled by $\Sigma_{\mathfrak{H}}$. Note that this separation of Hilbert space is not only for the matter sector, but for the entire reduced density matrix because every string operator $\Sigma_{\mathfrak{D}}$ with its edge configuration $\mathfrak{D} \subset A$ commutes with the patch operator projector,

$$[\Sigma_{\mathfrak{D}}, \hat{\Pi}_{\mathfrak{H}}^{(s)}] = 0 \quad \forall\mathfrak{D} \subset A.$$

For subsystems with two holes, the density matrix takes a similar form.

$$\rho_A = \frac{1}{2^{N^A-N_p^A-2}}\hat{\Pi}_A \underbrace{\left(\frac{\mathbb{1}+W_{\mathcal{L}_1}}{2}\right)\left(\frac{\mathbb{1}+W_{\mathcal{L}_2}}{2}\right)}_{\hat{\Pi}_{\mathcal{L}}} \sum_{[\mathfrak{D}]\subset A}\left(\begin{array}{l}\mathfrak{C}_{\mathfrak{D}}\Sigma_{\mathfrak{D}} + \mathfrak{C}_{\mathfrak{D}\ominus\mathfrak{H}_1}\Sigma_{\mathfrak{D}\ominus\mathfrak{H}_1} \\ \quad + \mathfrak{C}_{\mathfrak{D}\ominus\mathfrak{H}_2}\Sigma_{\mathfrak{D}\ominus\mathfrak{H}_2} + \mathfrak{C}_{\mathfrak{D}\ominus\mathfrak{H}_1\ominus\mathfrak{H}_2}\Sigma_{\mathfrak{D}\ominus\mathfrak{H}_1\ominus\mathfrak{H}_2}\end{array}\right).$$

The string operators that contain patch configurations and their corresponding coefficients can be written as the following forms.

$$\begin{cases} \mathfrak{C}_{\mathfrak{D}\ominus\mathfrak{H}_i}\Sigma_{\mathfrak{D}\ominus\mathfrak{H}_i} = \left(\dfrac{\varphi(\mathfrak{D},\mathfrak{H}_i)\mathfrak{C}_{\mathfrak{D}\ominus\mathfrak{H}_i}}{\mathfrak{C}_{\mathfrak{D}}}\right)\Sigma_{\mathfrak{H}_i}\mathfrak{C}_{\mathfrak{D}}\Sigma_{\mathfrak{D}} \equiv c_i^{\mathfrak{D}}\Sigma_{\mathfrak{H}_i}\mathfrak{C}_{\mathfrak{D}}\Sigma_{\mathfrak{D}} \\ \mathfrak{C}_{\mathfrak{D}\ominus\mathfrak{H}_1\ominus\mathfrak{H}_2}\Sigma_{\mathfrak{D}\ominus\mathfrak{H}_1\ominus\mathfrak{H}_2} = \left(\dfrac{\varphi(\mathfrak{H}_1,\mathfrak{H}_2\ominus\mathfrak{D})\varphi(\mathfrak{H}_2,\mathfrak{D})\mathfrak{C}_{\mathfrak{D}\ominus\mathfrak{H}_1\ominus\mathfrak{H}_2}}{\mathfrak{C}_{\mathfrak{D}}}\right)\Sigma_{\mathfrak{H}_1}\Sigma_{\mathfrak{H}_2}\mathfrak{C}_{\mathfrak{D}}\Sigma_{\mathfrak{D}} \equiv c_{12}^{\mathfrak{D}}\Sigma_{\mathfrak{H}_1}\Sigma_{\mathfrak{H}_2}\mathfrak{C}_{\mathfrak{D}}\Sigma_{\mathfrak{D}} \end{cases}$$

After that, we obtain coefficients $\mathbf{c}_{\mathfrak{D}}^T = (1, c_1^{\mathfrak{D}}, c_2^{\mathfrak{D}}, c_{12}^{\mathfrak{D}})$ for each configuration $\mathfrak{D}$ in the subsystem A as weights in front of the patch string operators. This way, we can separate the patch string operators from the string operators $\Sigma_{\mathfrak{D}}$ supported on the subsystem A. The patch string operators can be converted to a sum of projectors as follows,

$$\frac{1}{2^2}\left(\mathbb{1} + c_1^{\mathfrak{D}}\Sigma_{\mathfrak{H}_1} + c_2^{\mathfrak{D}}\Sigma_{\mathfrak{H}_2} + c_{12}^{\mathfrak{D}}\Sigma_{\mathfrak{H}_1}\Sigma_{\mathfrak{H}_2}\right) = \bigoplus_{\{s\}} p_{\mathfrak{D}}^{(\{s\})}\prod_{i=1}^{2}\left(\frac{\mathbb{1}+s_i\Sigma_{\mathfrak{H}_i}}{2}\right).$$

To achieve this, the coefficients $\mathbf{c}_{\mathfrak{D}}$ and the weight $p_{\mathfrak{D}}^{(\{s\})}$ in front of each projector must satisfy the following relations

$$\begin{cases} p_{\mathfrak{D}}^{(++)} + p_{\mathfrak{D}}^{(+-)} + p_{\mathfrak{D}}^{(-+)} + p_{\mathfrak{D}}^{(--)} = 1 \\ p_{\mathfrak{D}}^{(++)} + p_{\mathfrak{D}}^{(+-)} - p_{\mathfrak{D}}^{(-+)} - p_{\mathfrak{D}}^{(--)} = c_1^{\mathfrak{D}} \\ p_{\mathfrak{D}}^{(++)} - p_{\mathfrak{D}}^{(+-)} + p_{\mathfrak{D}}^{(-+)} - p_{\mathfrak{D}}^{(--)} = c_2^{\mathfrak{D}} \\ p_{\mathfrak{D}}^{(++)} - p_{\mathfrak{D}}^{(+-)} - p_{\mathfrak{D}}^{(-+)} + p_{\mathfrak{D}}^{(--)} = c_{12}^{\mathfrak{D}} \end{cases}$$

The first condition implies that the weights $p_{\mathfrak{D}}^{(\{s\})}$ can be interpreted as a probability distribution[1]. Writing the above relations in matrix form, it happens that the transform matrix is the Hadamard matrix.

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}\begin{pmatrix} p_{\mathfrak{D}}^{(++)} \\ p_{\mathfrak{D}}^{(+-)} \\ p_{\mathfrak{D}}^{(-+)} \\ p_{\mathfrak{D}}^{(--)} \end{pmatrix} = H_4\begin{pmatrix} p_{\mathfrak{D}}^{(++)} \\ p_{\mathfrak{D}}^{(+-)} \\ p_{\mathfrak{D}}^{(-+)} \\ p_{\mathfrak{D}}^{(--)} \end{pmatrix} = \begin{pmatrix} 1 \\ c_1^{\mathfrak{D}} \\ c_2^{\mathfrak{D}} \\ c_{12}^{\mathfrak{D}} \end{pmatrix}.$$

In general, this basis-changing trick applies to subsystems with an arbitrary number of holes, from which we obtain a generic relation

$$\mathbf{p}_{\mathfrak{D}} = \frac{1}{2^g}H_{2^g}\mathbf{c}_{\mathfrak{D}}$$

where $[\mathbf{p}_{\mathfrak{D}}]_i = p_{\mathfrak{D}}^{(\{s\}_i)}$ and $[\mathbf{c}_{\mathfrak{D}}]_j \propto \mathfrak{C}_{\mathfrak{D}\ominus\{\mathfrak{H}\}_j}/\mathfrak{C}_{\mathfrak{D}}$ are 2^g-dimensional vectors. With this decomposition, the reduced density matrix of a subsystem with an arbitrary number of holes can be written as

$$\rho_A = \frac{1}{2^{N^A - N_p^A - 2g}}\hat{\Pi}_A\hat{\Pi}_g\sum_{[\mathfrak{D}]\subset A}\mathfrak{C}_{\mathfrak{D}}\bigoplus_{\{s\}}p_{\mathfrak{D}}^{(\{s\})}\hat{\Pi}_{\mathfrak{H}}^{(\{s\})}\Sigma_{\mathfrak{D}}. \tag{1}$$

In the above formula, g represents the number of holes inside the subsystem (genus number), and $\{s\} \equiv \{s_1, s_2, \cdots, s_g\}$ is a set of eigenvalues of patch string operators $\Sigma_{\mathfrak{H}^{(n)}}$ covering different holes

[1] One more condition we need to check is whether each weight $p_{\mathfrak{D}}^{(\{s\})}$ is positive, but I currently have no proof.

$B^{(n)}$ that label different sectors. The projectors and the probability distribution are generalized as follows.

$$\begin{cases} \hat{\Pi}_g \equiv \prod_{n=1}^{g} \left(\frac{\mathbb{1} + W_{\mathcal{L}^{(n)}}}{2} \right) \\ \hat{\Pi}_{\mathfrak{H}}^{(\{s\})} \equiv \prod_{n=1}^{g} \left(\frac{\mathbb{1} + s_n \Sigma_{\mathfrak{H}^{(n)}}}{2} \right) \end{cases} \tag{2}$$

where $\mathcal{L}^{(n)}$ represents the non-contractible Wilson loop enclosing the n-th hole, and $\mathfrak{H}^{(n)}$ represents the patch configuration covering the n-th hole. The set $\{p_{\mathfrak{D}}^{(\{s\})}\}$ of each configuration class $[\mathfrak{D}]$ forms a probability distribution. The Hilbert space dimension of each sector can be computed straightforwardly

$$\mathrm{Tr}_A \left(\hat{\Pi}_A \hat{\Pi}_g \hat{\Pi}_{\mathfrak{H}}^{(\{s\})} \right) = 2^{N^A - N_p^A - 2g}.$$

We can further simplify the structure of the reduced density matrix to make it resemble the equivalent Majorana fermionic reduced density matrix. If this is doable, it may imply that the conventional matter-gauge separation picture in simply connected subsystems still applies for multiply-connected subsystems. First, we introduce the **projected string operators** as follows,

$$\hat{\mathcal{S}}_{\mathfrak{D}} \equiv \hat{\Pi}_A \hat{\Pi}_g \bigoplus_{\{s\}} p_{\mathfrak{D}}^{(\{s\})} \hat{\Pi}_{\mathfrak{H}}^{(\{s\})} \Sigma_{\mathfrak{D}}. \tag{3}$$

Note that the algebraic relation satisfied by the projected string operators is not strictly identical to that satisfied by the Majorana fermionic representation. **The multiplication of projected string operators does not form a closed algebra**.

$$\hat{\mathcal{S}}_{\mathfrak{D}} \hat{\mathcal{S}}_{\mathfrak{D}'} = \varphi(\mathfrak{D}, \mathfrak{D}') \hat{\Pi}_A \hat{\Pi}_g \bigoplus_{\{s\}} p_{\mathfrak{D}}^{(\{s\})} p_{\mathfrak{D}'}^{(\{s\})} \hat{\Pi}_{\mathfrak{H}}^{(\{s\})} \Sigma_{\mathfrak{D} \ominus \mathfrak{D}'}.$$

As shown above, multiplying two string operators no longer yields a string operator because of the probability weight.

With the projected string operators, the reduced density matrix can be organized into the following form that resembles the Majorana fermionic reduced density matrix of the equivalent Majorana fermionic system.

$$\rho_A = \frac{1}{2^{|\partial A|/2 - N_\partial^A}} \left[\frac{1}{2^{N^A/2}} \sum_{[\mathfrak{D}] \subset A} \mathfrak{C}_{\mathfrak{D}} \hat{\mathcal{S}}_{\mathfrak{D}} \right] \tag{4}$$

In the above equation, we have used the generalized geometric relation (19), and $N_\partial^A \equiv (1+g)$ is the number of disconnected boundaries of the subsystem A.

$$\begin{aligned} \mathrm{Tr}_A \rho_A^2 &= \frac{1}{2^{2\left(N^A - N_p^A - 2g\right)}} \sum_{[\mathfrak{D}],[\mathfrak{D}'] \subset A} \mathfrak{C}_{\mathfrak{D}} \mathfrak{C}_{\mathfrak{D}'} \mathrm{Tr}_A \left(\hat{\mathcal{S}}_{\mathfrak{D}} \hat{\mathcal{S}}_{\mathfrak{D}'} \right) \\ &= \frac{1}{2^{2\left(N^A - N_p^A - 2g\right)}} \sum_{[\mathfrak{D}] \subset A} \eta_{\mathfrak{D}} \mathfrak{C}_{\mathfrak{D}}^2 \sum_{\{s\}} \left(p_{\mathfrak{D}}^{(\{s\})} \right)^2 \mathrm{Tr}_A \left(\hat{\Pi}_A \hat{\Pi}_g \hat{\Pi}_{\mathfrak{H}}^{(\{s\})} \right) \\ &= \frac{1}{2^{N^A - N_p^A - 2g}} \sum_{[\mathfrak{D}] \subset A} \eta_{\mathfrak{D}} \mathfrak{C}_{\mathfrak{D}}^2 \sum_{\{s\}} \left(p_{\mathfrak{D}}^{(\{s\})} \right)^2 \leq \frac{1}{2^{N^A - N_p^A - 2g}} \sum_{[\mathfrak{D}] \subset A} \eta_{\mathfrak{D}} \mathfrak{C}_{\mathfrak{D}}^2 \end{aligned}$$

$$R_2\left(\rho_A\right) = -\ln \mathrm{Tr}_A \rho_A^2 \geq -\ln\left(\frac{1}{2^{N^A-N_p^A-2g}}\sum_{[\mathfrak{D}]\subset A}\eta_{\mathfrak{D}}\mathfrak{C}_{\mathfrak{D}}^2\right) = R_2\left(\rho_A^F\right) + \left(\frac{|\partial A|}{2} - N_\partial^A\right)\ln 2 \tag{5}$$

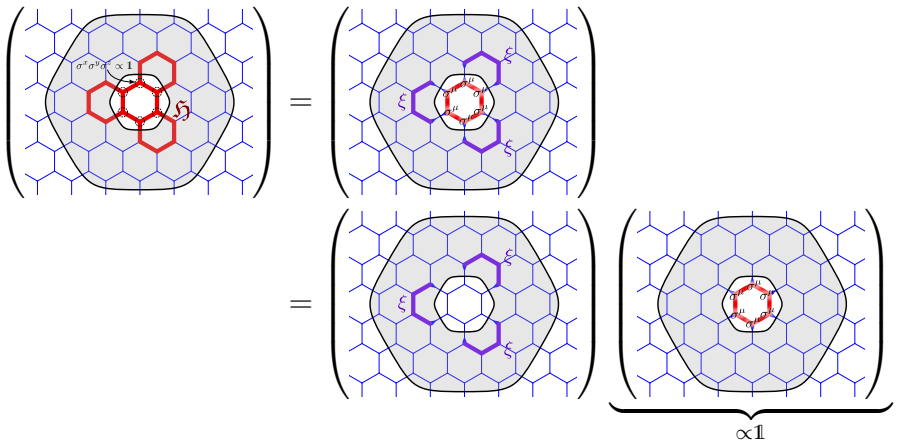
$$\tilde{p}^{(\{s\})} \equiv \mathrm{Tr}_A\left(\rho_A \hat{\Pi}_{\mathfrak{H}}^{(\{s\})}\right) = p_\varnothing^{(\{s\})} \tag{6}$$

$$\rho_A = \bigoplus_{\{s\}} \tilde{p}^{(\{s\})} \varrho_A^{(\{s\})} \tag{7}$$

$$\varrho_A^{(\{s\})} \equiv \frac{1}{2^{N^A-N_p^A-2g}}\hat{\Pi}_A\hat{\Pi}_g\hat{\Pi}_{\mathfrak{H}}^{(\{s\})}\sum_{[\mathfrak{D}]\subset A}\frac{p_{\mathfrak{D}}^{(\{s\})}}{p_\varnothing^{(\{s\})}}\mathfrak{C}_{\mathfrak{D}}\Sigma_{\mathfrak{D}}$$

2 Patch String Operators Interpreted as Parity of Residual Electric Fields

The physical meaning of the patch string operator $\Sigma_{\mathfrak{H}}$ can be interpreted in the other way. The string operator $\Sigma_{\mathfrak{H}}$ of the patch configuration can be decomposed into the multiplication of all the ξ operators along the interior boundary and the Pauli spin operators that represent the residual electric field inside the hole along the boundary attached to the string operators covering the hole. This decomposition is pictorially illustrated as follows.

Σ 

$$= \left(\prod_{\hat{\xi}\in\partial B}\hat{\xi}\right)\underbrace{\left(\prod_{\langle ij\rangle\in B}\sigma_i^{\alpha_{ij}}\sigma_j^{\alpha_{ij}}\right)\left(\prod_{i\in\partial B}\sigma_i^{\mu_i}\right)}_{\propto\mathbb{1}} \propto \prod_{\hat{\xi}\in\partial B}\hat{\xi} \tag{8}$$

From this result, we see that the string operator $\Sigma_{\mathfrak{H}}$ of the patch configuration is the product of all the ξ-operators along the interior boundary. Accordingly, the projector $\hat{\Pi}_{\mathfrak{H}}$ can be redefined as follows,

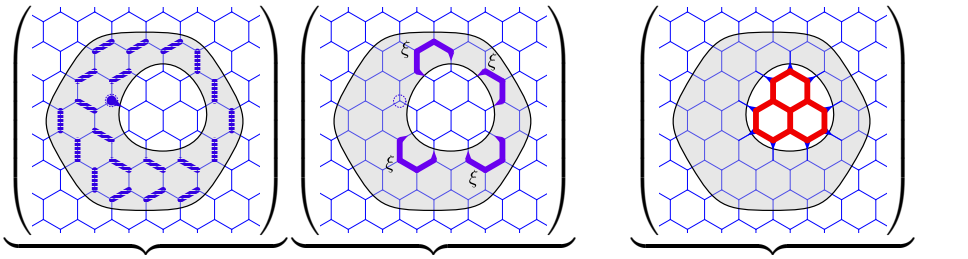
$$\hat{\Pi}_{\mathfrak{H}}^{(\{s\})} = \prod_{n=1}^{g}\frac{1}{2}\left(\mathbb{1} + s_n\prod_{\hat{\xi}\in\partial B^{(n)}}\hat{\xi}\right). \tag{9}$$

This result clearly indicates that **the patch sector actually corresponds to the parity of electric fields surrounding each hole** $B^{(n)}$.

Note that **we don't have to consider this pattern in a simply connected subsystem** since we have already included the configurations satisfying $[\mathfrak{D} \ominus \mathfrak{D}_Z] \subset A$ when constructing the reduced density matrix. The string operator defined as $\prod_{\hat{\xi}\in\partial A}\hat{\xi}$ in a simply connected region A is equivalent to the combination of the product of all the dimers lying outside A, the dimers crossing the boundary ∂A, and the gauge links forming ξ-operators along the boundary. This configuration is taken into account by $[\mathfrak{D} \ominus \mathfrak{D}_Z] \subset A$. For a similar reason, **we should not include the outer boundary either**, because otherwise the reduced density matrix ρ_A will contain the component $\prod_{\hat{\xi}\in\partial A}\hat{\xi}$ consisting of all the ξ operators along all the boundaries of the region A as mentioned earlier. Such an operator is equivalent to multiplying all the Majorana fermions lying outside the subsystem A, and which has already been taken into account in ρ_A since such a string operator satisfies the condition $[\mathfrak{D} \ominus \mathfrak{D}_Z] \subset A$.

Following the same pattern, we can define the proper $\Sigma_{\mathfrak{H}}$ for subsystems whose **inner** boundary has **odd** length and **outer** boundary has **even** length.

$$\Sigma_{\mathfrak{H}} \equiv$$



$$=\hat{\Xi} \qquad =\prod_{\hat{\xi}\in\partial B}\hat{\xi} \qquad =\left(\prod_{\langle ij\rangle\in B}\sigma_i^{\alpha_{ij}}\sigma_j^{\alpha_{ij}}\right)\left(\prod_{i\in\partial B}\sigma_i^{\mu_i}\right)\propto\mathbb{1}$$

$$= \hat{\Xi}\left(\prod_{\hat{\xi}\in\partial B}\hat{\xi}\right)\underbrace{\left(\prod_{\langle ij\rangle\in B}\sigma_i^{\alpha_{ij}}\sigma_j^{\alpha_{ij}}\right)\left(\prod_{i\in\partial B}\sigma_i^{\mu_i}\right)}_{\propto\mathbb{1}} \propto \hat{\Xi}\prod_{\hat{\xi}\in\partial B}\hat{\xi} \tag{10}$$

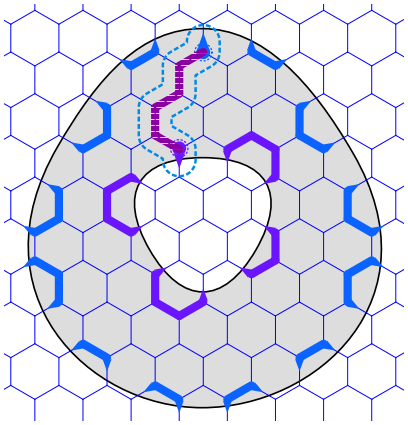


Figure 2: An unsolved problem is the pattern of patch sectors of subsystems with both their inner and outer boundaries having odd length. In this case, there is no well-defined Ξ operator, but the ξ operators surrounding the inner boundary cannot form a valid patch configuration either. A possible resolution is to connect the remaining sites at the inner and outer boundaries. One example is the Wilson operator enclosed by the dashed frame in the above illustration. However, this method also requires the ξ operators along the outer boundary, which should not be included in this framework.

3 Extensive Degeneracy in the Entanglement Spectrum

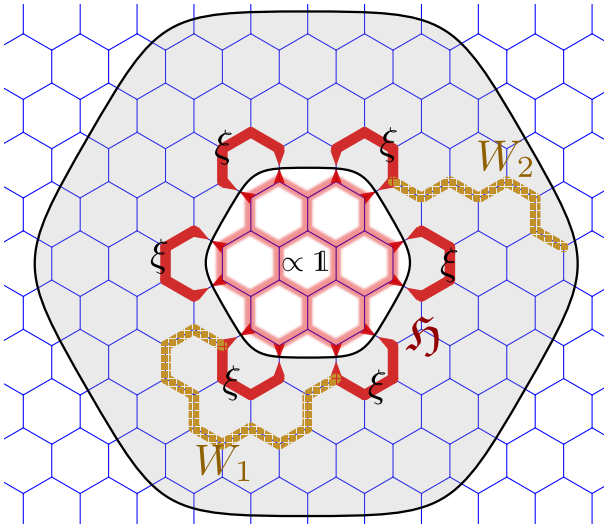


Figure 3: The non-contractible Wilson operator W_2 with two endpoints attached to different boundaries anti-commutes with the patch operator $\Sigma_{\mathfrak{H}}$ covering the interior hole, which is proportional to the product of all the ξ operators surrounding the inner boundary, while the non-contractible Wilson operator W_1 with both endpoints attached to the patch commutes with $\Sigma_{\mathfrak{H}}$ as well as other string operators supported within the subsystem A, hence commutes with the reduced density matrix ρ_A.

The non-contractible Wilson operators connecting two electric-field operators fall into two types. One type, denoted as $\mathfrak{W}_s$, has both endpoints attached to the same boundary, while the other, denoted as $\mathfrak{W}_d$, has endpoints attached to different boundaries. As illustrated in Fig. 1(b), the non-contractible Wilson operator W_1 is of type $\mathfrak{W}_s$, while the other Wilson operator W_2 is of type $\mathfrak{W}_d$.

$$W_2 \hat{\Pi}_{\mathfrak{H}}^{(\pm)} W_2^\dagger = \hat{\Pi}_{\mathfrak{H}}^{(\mp)}$$

Consequently, the non-contractible Wilson operators in $\mathfrak{W}_d$ do not commute with the reduced density matrix ρ_A, so they are not proper unitary operators for proving that the subblocks are identical. In contrast, the operators in $\mathfrak{W}_s$ let the patch projectors remain intact. Thus, they commute with the density matrix and become the only proper operators for our purpose. Our arguments lead to the conclusion: **the non-contracted Wilson operators that relate different subblocks are those with both endpoints attached to the same boundary**. As a result, the degree of degeneracy of the entanglement spectrum is equal to the number L of different $\{\xi; \Xi\}$ configurations divided by two to the power of the number of disconnected boundaries due to the conservation of residual electric field parity along each inner boundary, which is equal to

$$\text{Degree of degeneracy} = \frac{2^L}{2^{N_\partial^A}} = 2^{L-N_\partial^A}$$

If all the boundaries have even length, the number of residual electric field configurations L reduces to $L = |\partial A|/2$, which reproduces the conventional result

$$\text{Degree of degeneracy} = 2^{|\partial A|/2 - N_\partial^A} = \exp\left(S_G^A\right) \tag{11}$$

According to the above discussion, each subblock corresponds to a specific configuration of ξ's eigenvalues along the boundary, but we also know that the Wilson operators can relate only blocks

with the same ξ-parity on each boundary. Therefore, the projector onto the subblock represented by the configuration $\{\xi\}$ reads

$$\hat{\Pi}_{[\{\xi\}]} \equiv \bigoplus_{\{\xi\}\in[\{\xi\}]} \prod_{\substack{\hat{\xi}_i\in\partial A \\ \xi_i\in\{\xi\}}} \left(\frac{\mathbb{1}+\xi_i\hat{\xi}_i}{2}\right). \tag{12}$$

In the above definition, $[\{\xi\}]$ is a group of 2^g configurations represented by $\{\xi\}$ with different ξ-parity on each boundary. For example, if the subsystem A has one inner hole, that is to say, it has two disconnected boundaries, then a class $[\{\xi\}]$ contains $2^2 = 4$ sets of ξ-configurations

$$\{\xi\}_{++} \equiv \{\xi\}\,,\ \{\xi\}_{+-}\,,\ \{\xi\}_{-+}\,,\ \{\xi\}_{--}\,,$$

where $\{\xi\}_{s_1s_2}$ represents the ξ-configuration with the outer ξ-parity being s_1 and the inner ξ-parity being s_2. There is not necessarily a relationship among $\{\xi\}_{s_1s_2}$ in the same class $[\{\xi\}]$. We did this grouping merely to divide the reduced density matrix into identical blocks. We can further separate the class $[\{\xi\}]$ into two parts

$$[\{\xi\}] = [\{\xi\}]^+ \cup [\{\xi\}]^-,$$

where $[\{\xi\}]^\pm$ denotes the subclass of sets with the total parity even/odd, respectively. Take a subsystem with one hole, for instance. The subclasses are

$$\begin{cases} [\{\xi\}]^+ = \left\{\{\xi\}_{++}, \{\xi\}_{--}\right\} \\ [\{\xi\}]^- = \left\{\{\xi\}_{+-}, \{\xi\}_{-+}\right\} \end{cases}$$

With these structures, the reduced density matrix can be written as a direct sum of identical subblocks, as argued earlier.

$$\rho_A = \bigoplus_{[\{\xi\}]} p_{[\{\xi\}]}\varrho_A^{[\{\xi\}]} \tag{13}$$

The probability of finding the state in a certain block is a constant.

$$\begin{aligned} p_{[\{\xi\}]} &= \mathrm{Tr}_A\left(\rho_A\hat{\Pi}_{[\{\xi\}]}\right) \\ &= \frac{1}{2^{|\partial A|/2}} \sum_{\{\xi\}\in[\{\xi\}]} \mathrm{Tr}_A\left[\rho_A\left(\mathbb{1} + \prod_{\xi_i\in\{\xi\}}\xi_i \prod_{\hat{\xi}_j\in\partial A}\hat{\xi}_j\right)\right] \\ &= \frac{1}{2^{|\partial A|/2}}\left[\sum_{\{\xi\}\in[\{\xi\}]^+}\left(1+e^{i\phi_A}\left\langle\prod_{i\in A}c_i\right\rangle\right) + \sum_{\{\xi\}\in[\{\xi\}]^-}\left(1-e^{i\phi_A}\left\langle\prod_{i\in A}c_i\right\rangle\right)\right] \\ &= \frac{1}{2^{|\partial A|/2-N_\partial^A}}. \end{aligned}$$

Although the probability distribution is as trivial as demonstrated above, solving the finer entan-

glement spectrum of each block is not an easy task.

$$\begin{aligned}\varrho_A^{[\{\xi\}]} &\equiv \hat{\Pi}_{[\{\xi\}]}\rho_A \\ &= \sum_{[\mathfrak{D}]\subset A}\sum_{\{s\}} p_{\mathfrak{D}}^{(\{s\})} \sum_{\{\xi\}\in[\{\xi\}]} \prod_{\substack{\hat{\xi}_i\in\partial A \\ \xi_i\in\{\xi\}}} \left(\frac{\mathbb{1}+\xi_i\hat{\xi}_i}{2}\right) \prod_{n=1}^{g}\left(\frac{\mathbb{1}+s_n\Sigma_{\mathfrak{H}^{(n)}}}{2}\right) \frac{1}{2^{N^A-N_p^A-2g}}\hat{\Pi}_A\hat{\Pi}_g\mathfrak{C}_{\mathfrak{D}}\Sigma_{\mathfrak{D}} \\ &= \text{How should we proceed?}\end{aligned}$$

As long as we know the entanglement spectrum of each subblock, we can compute the entanglement entropy as we did previously.

$$S_A = \sum_{[\{\xi\}]} p_{[\{\xi\}]} S\left(\varrho_A^{[\{\xi\}]}\right) - \sum_{[\{\xi\}]} p_{[\{\xi\}]} \ln p_{[\{\xi\}]} = S\left(\varrho_A\right) + \left(\frac{|\partial A|}{2} - N_\partial^A\right)\ln 2 \tag{14}$$

4 Electric Field Parity, 't Hooft Loops, and Subsystem Fermion Parity

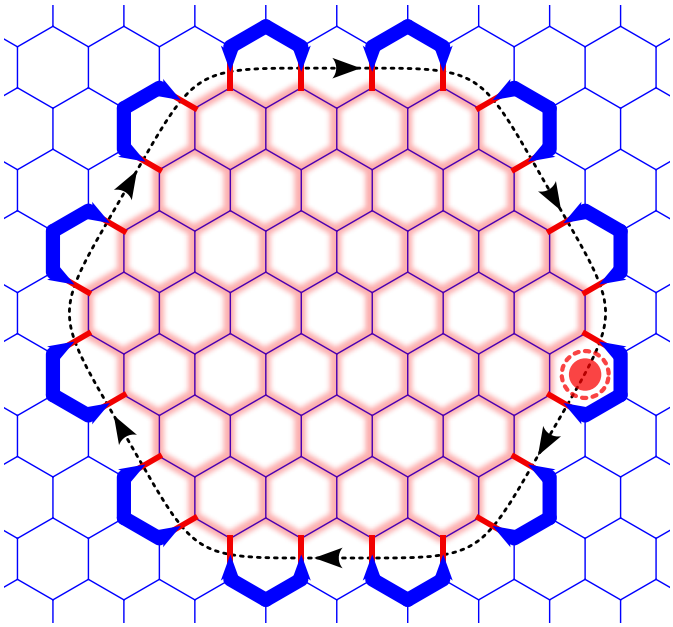


Figure 4: In the Kitaev model, assembling ξ-operators along a closed loop represents a 't Hooft loop of the $\mathbb{Z}_2$ gauge theory, which can be understood as the path of a $\mathbb{Z}_2$ gauge flux (vison) moving along the closed loop and eventually being annihilated by its partner created at the same spot at the very beginning. It is also equivalent to the subsystem fermion parity of the region enclosed by the 't Hooft loop.

Aside from the Wilson loops, the 't Hooft loops should also be discussed since the low-energy effective theory of the Kitaev model is a TQFT. The 't Hooft loop in this note is operationally defined as the operator that creates a pair of $\mathbb{Z}_2$ gauge fluxes (visons) and then moves one of the visons around a loop until it gets back to the initial spot to annihilate its partner vison. An example of a 't Hooft loop is illustrated in Fig. 4 as represented by a dashed closed loop crossing a series of bonds at which the $\mathbb{Z}_2$ gauge fields are flipped by the electric fields. From the gauge theory perspective, the 't Hooft operator $\tilde{T}_{\mathcal{C}}$ along a closed loop $\mathcal{C}$ in the Kitaev model can be

defined as follows

$$\tilde{T}_{\mathcal{C}} = \prod_{i\in\mathcal{C}} b_i^{\mu_i} = \exp\left(i\pi \oint_{\mathcal{C}} E_\perp \mathrm{d}\ell\right),$$

where the electric fields are provided by the b-Majorana fermionic operators along the loop $\mathcal{C}$. The above interpretation, combined with our discussions on the relationship between the patch string operators and electric fields, leads to a picture that the 't Hooft loop, which is equivalent to the electric field parity along the loop, is also related to the subsystem fermion parity of the region enclosed by the loop. Such a relationship is formulated mathematically as follows,

$$\tilde{T}_{\partial\Gamma} = \exp\left(i\pi \oint_{\partial\Gamma} E_\perp \mathrm{d}\ell\right) \leftrightsquigarrow \prod_{\hat{\xi}\in\partial\Gamma} \hat{\xi} = \Sigma_{\mathfrak{H}} \leftrightsquigarrow \prod_{i\in\Gamma} c_i \propto (-1)^{F_\Gamma}\ ; \quad F_\Gamma = \sum_{i\in\Gamma} n_i, \tag{15}$$

where $E_\perp$ represents the electric field components perpendicular to the 't Hooft loop direction, and $(-1)^{F_\Gamma}$ stands for the subsystem fermion parity of the region Γ. According to the divergence theorem, the relationship between the electric fields along the loop $\partial\Gamma$ and the subsystem fermion parity can be translated to the Gauss law of the gauge theory,

$$\oint_{\partial\Gamma} E_\perp \mathrm{d}\ell = \sum_{i\in\Gamma} n_i \Rightarrow \boldsymbol{\nabla}\cdot E = \rho.$$

This is a generalization of the elementary Gauss law $b_i^x b_i^y b_i^z c_i = \mathbb{1}$ which forms an elementary triangular loop enclosing site i. In conclusion, the Gauss law of the gauge theory in the Kitaev model implies the equivalence between the 't Hooft loop and the subsystem fermion parity. We can accordingly rewrite the patch projectors in terms of 't Hooft loop operators enclosing each hole inside the multiply-connected subsystem.

$$\hat{\Pi}_{\mathfrak{H}}^{(\{s\})} = \prod_{n=1}^{g}\left(\frac{\mathbb{1}+s_n\tilde{T}_{\partial B^{(n)}}}{2}\right) = \prod_{n=1}^{g}\left(\frac{\mathbb{1}+s_n(-1)^{F_n}}{2}\right). \tag{16}$$

In Ref.[1], the topological entanglement entropy is interpreted as a consequence of **spontaneously broken 't Hooft 1-form symmetry**. Formally, the state $|\psi\rangle$ breaks the 't Hooft 1-form symmetry if $T|\psi\rangle$ is orthogonal to $|\psi\rangle$. In the Kitaev model, the 't Hooft loop is not a symmetry since it doesn't commute with the Hamiltonian. However, it does serve as a source to produce the topological entanglement entropy as discussed in previous sections.

5 Disconnected Regions and Complement Picture

In this section, we provide another insight into how the entanglement spectrum depends on the subsystem topology. We will show that it arises from the **difference in fermion-parity behavior between the physical Kitaev model and its equivalent Majorana system**.

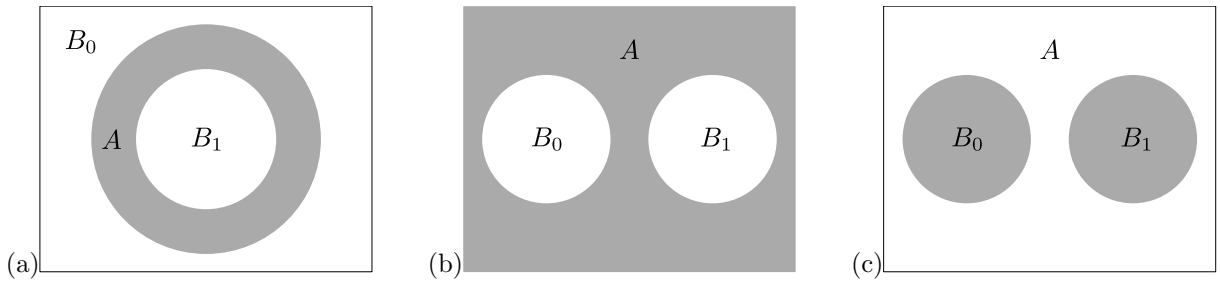


Figure 5: An annulus is topologically equivalent to creating two vacancies in the universe, and its complement is composed of two disconnected subregions. Suppose the state defined on the universe (the entire system) is pure; then the entanglement entropy of the subsystem A should be equivalent to the entanglement entropy of the union of two holes $\overline{A} = B_0 \cup B_1$.

As illustrated in Fig. 5, the entanglement entropy of an annulus-shaped subsystem is topologically equivalent to the entanglement entropy of a subsystem composed of two disconnected regions if the entire state is pure. Mathematically, we have

$$S_A = S_{\overline{A}} = S_{B_0 \cup B_1}$$

Furthermore, the entanglement spectrum of the original problem and its complement are also identical. This is what I call the *complement method*, in which I switch the environment and the system such that the original geometry is converted to a union of disconnected subsystems. Below, we discuss the entanglement entropy $S_{B_0 \cup B_1}$ of the Kitaev model for the subsystem $B_0 \cup B_1$ illustrated in Fig. 5(c).

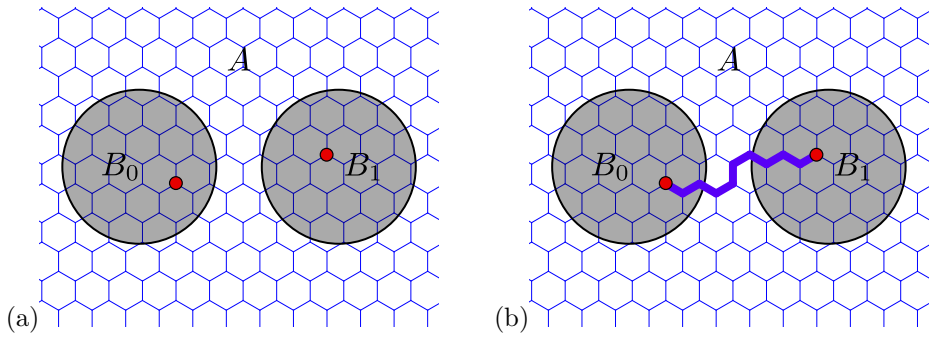


Figure 6: (a) Creating two Majorana fermions separately in two disconnected regions in a pure Majorana system is valid. (b) However, in the Kitaev model, creating two fermions in two separated regions requires a string across them, and this string operator is forbidden to appear in the complement reduced density matrix $\rho_{\overline{A}}$ of the subsystem $\overline{A} = B_0 \cup B_1$.

We note that the Kitaev model is not completely equivalent to its Majorana counterpart because the fermion parity is realized by Pauli strings under a stricter condition. As illustrated in Fig. 6(a), a basis $c_1 c_2$ is allowed to appear in the fermionic density matrix $\rho^F_{B_0 \cup B_1}$ even if c_1 and c_2 are located in two disconnected subregions, respectively. However, it is not the case in the Kitaev model, where realizing two fermions requires a non-local string operator $\Sigma_{\overline{12}}$ connecting the two fermions. However, it is inevitable for the string $\Sigma_{\overline{12}}$ to cross the environment A if the two Majorana fermions c_1 and c_2 are located in two disconnected subregions, respectively. As a consequence, this type of string operator doesn't appear in the physical reduced density matrix $\rho_{B_0 \cup B_1}$. In other words, **a fermion-parity-odd term is not allowed in each simply connected subregion, even**

if the overall fermion-parity across the entire subsystem of such a term is even. This constraint results in the *physical*[2] Majorana reduced density matrix for subsystems composed of disconnected components.

$$\rho_{\overline{A}}^{F(\text{phys})} = \frac{1}{2^{N^{\overline{A}}/2}} \sum_{\text{EP}(\mathfrak{D})\subset\overline{A}}{}' \mathfrak{C}_{\mathfrak{D}} \mathcal{M}_{\mathfrak{D}}. \tag{17}$$

The primed sum in Eq. (17) sums over configurations satisfying the condition that there are an even number of Majorana fermions in each simply-connected component of $\overline{A}$. The constraint on subsystem fermion parity naturally leads to the fact that the subsystem fermion parity operator of each component B_n commutes with the reduced density matrix $\rho_{\overline{A}}^{F(\text{phys})}$

$$[(-1)^{F_n}, \mathcal{M}_{\mathfrak{D}}] = 0 \Longrightarrow [(-1)^{F_n}, \rho_{\overline{A}}^{F(\text{phys})}] = 0 \quad \forall n$$

The subsystem-wise fermionic conservation is equivalent to the conservation of ξ-parity on each boundary of A in the original problem, as we showed in the previous section that the ξ-parity can be interpreted as the subsystem fermion parity due to Gauss' law of the gauge theory. The conservation of subsystem fermion parity implies that the degree of degeneracy in the entanglement spectrum also satisfies the prediction given by the picture where open Wilson operators serve as unitary transforms connecting the subblocks labeled by the ξ-configurations. Similarly, we can still define the ξ-operators along each separated boundary. In principle, the ξ-operators defined along the boundaries a $\left|\partial\overline{A}\right|/2$-fold degeneracy. However, the conservation of subsystem fermion parity reduces the degree of degeneracy[3]. Because $\overline{A}$ has $N_\partial^{\overline{A}}$ separated components, the reduction in the degree of degeneracy is therefore $2^{N_\partial^{\overline{A}}}$. This results in the final degree of degeneracy in the entanglement spectrum

$$\text{Degree of degeneracy} = \frac{2^{\left|\partial\overline{A}\right|/2}}{2^{N_\partial^{\overline{A}}}} = 2^{|\partial A|/2 - N_\partial^A},$$

in agreement with the analysis (11), where we have used $|\partial A| = \left|\partial\overline{A}\right|$ and $N_\partial^A = N_\partial^{\overline{A}}$.

Now we return to the fundamentals of the complement trick. The equivalence between the two complement pictures relies on the fact that reversing the directions of the residual electric fields leaves the physics intact, because the ξ-operators in the **original problem** point **inward** to the hole regions, while the ξ-operators in the **complement picture** point **outward** from the hole regions. The equivalence between these two definitions of ξ-operators is based on the following equation, which says that the inward ξ-operators group is equivalent to the outward one up to some Wilson loops.

$$(-1)^{F_n} \leftrightsquigarrow \prod_{\hat{\xi}\in\partial B_n(\text{out})} \hat{\xi} = (\text{up to some Wilson loops}) \prod_{\hat{\xi}\in\partial B_n(\text{in})} \hat{\xi}. \tag{18}$$

An illustration of an example is given in Fig. (7).

[2]Here, the term *physical* means the Majorana fermionic state that satisfies the fermion parity constraint discussed previously. Not to be confused with the state in the physical spin representation.

[3]The reduction in the degeneracy can also be explained by the usual picture: A single open Wilson operator can only be defined inside a certain connected subregion. Therefore, it flips an even number of ξ-eigenvalues on the same boundary at the same time, and the ξ-parity of that subregion remains unchanged.

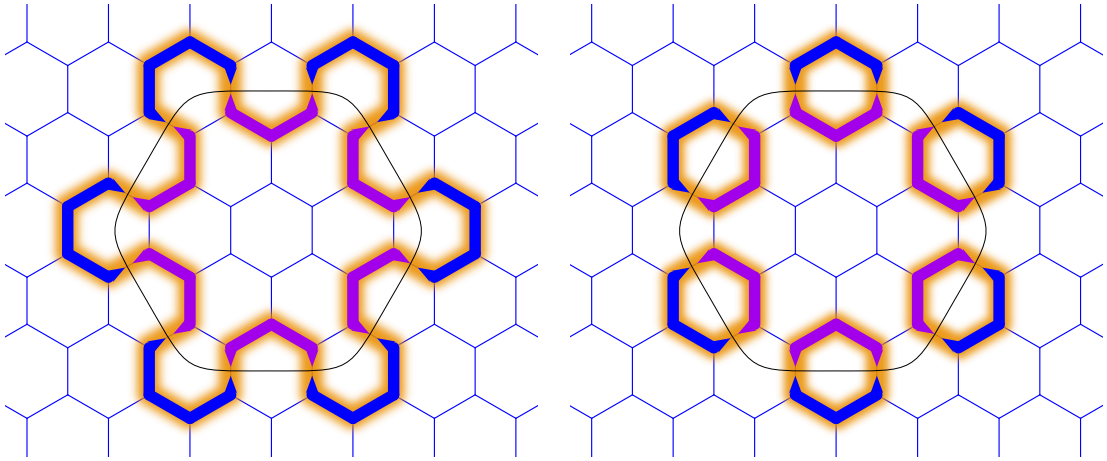


Figure 7: A pictorial illustration of Eq. (18). Combining the ξ-operators pointing a certain region inwards and those pointing outwards results in a Wilson loop (or an assembly of Wilson loops) around the boundary of the region. In other words, the combination of inner ξ-operators is equivalent to the combination of outer ξ-operators in a 1-form symmetry sense. The thick dentate segments represent ξ-operators, and the blurred loops represent Wilson loops.

6 Magic of Non-Simply Connected Subsystems

The reduced density matrix of a subsystem with a hole inside takes the following form

$$\rho_A = \frac{1}{2^{N^A - N_p^A - 2g}} \hat{\Pi}_A \hat{\Pi}_{\mathcal{L}} \sum_{[\mathfrak{D}] \subset A} \left(\frac{1}{2^g} \sum_{\{\mathfrak{H}\}} \mathfrak{C}_{\mathfrak{D} \ominus \{\mathfrak{H}\}} \Sigma_{\mathfrak{D} \ominus \{\mathfrak{H}\}} \right).$$

The summation of the expectation value of Pauli strings to the power four is then

$$\frac{1}{2^{N^A}} \sum_{\mathsf{S} \in \tilde{\mathcal{P}}_{N^A}} \left[\mathrm{Tr}_A \left(\rho_A \mathsf{S} \right) \right]^4 = \frac{1}{2^{N^A - N_p^A - 2g}} \sum_{[\mathfrak{D}] \subset A} \left[\frac{1}{2^g} \sum_{\{\mathfrak{H}\}} \left(\frac{\mathfrak{C}_{\mathfrak{D} \ominus \{\mathfrak{H}\}}}{\mathfrak{C}_{\mathfrak{D}}} \right)^4 \right] \mathfrak{C}_{\mathfrak{D}}^4$$

Because $\mathfrak{C}_{\mathfrak{D}}^2 \geq \mathfrak{C}_{\mathfrak{D} \ominus \{\mathfrak{H}\}}^2$, the series in the square bracket is less than unity,

$$\frac{1}{2^g} \sum_{\{\mathfrak{H}\}} \left(\frac{\mathfrak{C}_{\mathfrak{D} \ominus \{\mathfrak{H}\}}}{\mathfrak{C}_{\mathfrak{D}}} \right)^4 \leq \frac{1}{2^g} \sum_{n=1}^{2^g} = 1.$$

Therefore, the magic part of the subsystem 2-SRE is greater than that computed by the Majorana state with the topological boundary correction,

$$-\ln \left[\frac{1}{2^{N^A}} \sum_{\mathsf{S} \in \tilde{\mathcal{P}}_{N^A}} \left[\mathrm{Tr}_A \left(\rho_A \mathsf{S} \right) \right]^4 \right] \geq -\ln \left(\frac{1}{2^{N^A - N_p^A - 2g}} \sum_{[\mathfrak{D}] \subset A} \mathfrak{C}_{\mathfrak{D}}^4 \right).$$

A Generalized Geometric Relation

This formula can be generalized to multiply-connected regions. Suppose a connected region has a genus number g, and the n-th hole surrounded by the connected region A is denoted by $B^{(n)}$ such that $A \cap B^{(n)} = \varnothing$ and $A \cup \left(\bigcup_{n=1}^{g} B^{(n)} \right)$ is simply-connected. Denoting by $\ell_{\text{out}} =$

$\left|\partial\left(A\cup\left(\bigcup_{n=1}^{g}B^{(n)}\right)\right)\right|$ the number of crossing edges of the outer boundary, and by $\ell_{\text{in}}^{(n)}=\left|\partial B^{(n)}\right|$ the length of the n-th inner boundary, we have

$$\begin{cases}\ell_{\text{out}}=N^{A}-2N_{p}^{A}+\sum_{n=1}^{g}\left(N^{B^{(n)}}-2N_{p}^{B^{(n)}}-2N_{p}^{\partial B^{(n)}}\right)+2\\ \ell_{\text{in}}^{(n)}=N^{B^{(n)}}-2N_{p}^{B^{(n)}}+2\end{cases}$$

where $N_{p}^{\partial(n)}$ is the number of plaquettes at the interface between A and B. It is obvious that $N_{p}^{\partial(n)}=\ell_{\text{in}}^{(n)}$. Thus, we have

$$\ell_{\text{out}}=N^{A}-2N_{p}^{A}-\sum_{n=1}^{g}\ell_{\text{in}}^{(n)}-2g+2$$

At the same time, we know that the length of the boundary of the region A is the sum of the outer boundary and all inner boundaries

$$|\partial A|=\ell_{\text{out}}+\sum_{n=1}^{g}\ell_{\text{in}}^{(n)}$$

Combining this relation with the result we obtained gives a generalized geometric relation:

$$|\partial A|=N^{A}-2N_{p}^{A}+2(1-g) \tag{19}$$

The above relation implies that the number of lattice sites N^{A} determines whether the total boundary length $|\partial A|$ is even or odd.

$$N^{A}=|\partial A| \mod 2$$

References

[1] W.-T. Xu, T. Rakovszky, M. Knap, and F. Pollmann, *Entanglement Properties of Gauge Theories from Higher-Form Symmetries*, Phys. Rev. X **15**, 011001 (2025).